Linguistic Signals and Structural Biases in Headline-Level Sarcasm Detection

A0271934Y, A0273436B, A0277540B, A0273402N, A0300661U
Group 18
Mentored by Hejin Wang
{e1121337, e1122839, e1138462, e1122805, e1355027}@u.nus.edu

Abstract

We study sarcasm detection in news headlines as a binary classification task, focusing on *linguistic signals* over full-document or context-dependent interpretation. To support this, we construct a large deduplicated corpus and introduce a *controlled diagnostic validation set* that filters out instances requiring extralinguistic knowledge. We evaluate transformer-based models, namely BERT, RoBERTa, and DeBERTa under successive modifications of fine-tuning, hyperparameter tuning, and targeted data addition to the training pipeline. Results show that such modifications yield consistent improvements, though *excessive data addition can introduce noise and degrade performance*. Error analysis reveals linguistic biases, including spurious correlations with scientific terminology, political domains, and failures in identifying *tautological sarcasm*. Overall, our findings suggest that sarcasm detection is primarily constrained by *linguistic ambiguity and dataset artifacts*, rather than model capacity alone.

1 Introduction

The rapid proliferation of online news and social media has fundamentally transformed how information is consumed and disseminated. However, this accessibility has also amplified the spread of misinformation and disinformation, which the World Economic Forum identifies as a major short-term global risk in its 2025 report (World Economic Forum, 2025). A particularly subtle contributor to this issue is the misinterpretation of *satirical content* as factual news. Articles originating from satirical outlets, such as *The Onion*, are frequently detached from their original context and re-shared across platforms, where they may be mistaken for legitimate reporting.

Detecting sarcasm in text is a challenging problem in natural language processing (*NLP*), as it requires understanding implicit meaning, tone, and contextual cues that are often absent from surface-level lexical features (Mishra et al., 2016). Unlike traditional text classification tasks, *sarcasm detection* demands deeper semantic interpretation, making for a compelling and non-trivial problem.

In this project, we address the task of *sarcasm detection at the headline level*, rather than full-document satire classification or fake news detection. In modern digital environments, users frequently consume information in a highly time-constrained manner, often scanning through headlines without engaging with the full article. As a result, headlines play a disproportionately influential role in shaping readers' perceptions and beliefs.

Accordingly, we do not aim to determine the factual accuracy of a claim or classify headlines as misinformation. Instead, given a news headline as input, the model aims to predict a binary label indicating whether the headline is sarcastic or not. This scoped focus allows us to isolate the *linguistic challenges of sarcasm* – particularly in short, context-limited text – while maintaining relevance to the larger problem of information reliability. This formulation will focus on *linguistic signals* in the text, excluding *pragmatic interpretations* from the task definition.

2 Related Work / Background

Sarcasm detection has been widely studied as a challenging NLP task, with early work highlighting that sarcasm often cannot be reliably detected using surface-level lexical features or sentiment features alone, as it frequently depends on *semantic incongruity* between literal expression and intended meaning (Joshi et al., 2016). This has motivated neural approaches that leverage large-scale supervised learning to capture deeper semantic representations rather than surface-level sentiment cues. However, sarcasm remains particularly difficult in journalism settings due to the absence of non-verbal or contextual cues that are typically essential in interpreting sarcasm in human communication.

The News Headlines Dataset for Sarcasm Detection (NHDSD) is a prominent benchmark in this space, constructed by pairing headlines from a satirical publication (*The Onion*) and a mainstream news outlet (*HuffPost*) to enable scalable training (Misra and Arora, 2023). However, subsequent work has shown that sarcasm detection models trained on such single-source benchmarks fail to generalise across datasets with different styles, domains, or label sources, indicating that models often learn dataset-specific shortcuts rather than transferable signals of sarcasm (Jang and Frassinelli, 2024).

More recently, transformer-based models such as BERT, RoBERTa and DeBERTa have become standard baselines in sarcasm detection due to their ability to encode contextual representations. Comparative studies in related domains, particularly social media texts, show that these models consistently outperform traditional approaches, with DeBERTa often achieving the strongest performance (Jiménez Castro et al., 2026). However, two limitations persist. Models trained on a single benchmark often fail to generalise across datasets, suggesting sensitivity to *dataset artifacts* rather than transferrable sarcasm signals (Jang and Frassinelli, 2024); and sarcasm frequently depends on world knowledge or pragmatic reasoning outside the text itself (Joshi et al., 2016).

In this work, we build on prior sarcasm detection research at two levels. At the *dataset level*, we address concerns that single-source benchmarks such as NHDSD encourage shortcut learning (Jang and Frassinelli, 2024) by constructing a multi-source deduplicated corpus that extends NHDSD with additional satirical and non-satirical headlines. At the *model level*, we move beyond aggregate performance comparisons of transformer architectures toward a diagnostic analysis that isolates *which* linguistic and structural features drive residual errors, enabling a more rigorous assessment of whether models are learning *linguistic signals of sarcasm* or relying on dataset-specific shortcuts and contextual artifacts.

3 Corpus Analysis and Methods

3.1 Data Collection

To support robust sarcasm detection, we constructed an expanded corpus by aggregating data from multiple sources, containing both satirical and non-satirical news headlines. In addition to the provided dataset, we incorporated four external datasets to improve diversity and scale of data points. The URLs are in Appendix A, Table 5.

- The *Onion News* dataset from Hugging Face, containing articles and headlines from *The Onion*.
- Kaggle dataset featuring unusual, satirical-sounding headlines from Reddit's *r/NotTheOnion*, which we filtered to specifically analyze real-world news articles published by the Huffington Post.
- Kaggle dataset of *The Onion*'s satirical headlines, covering a wide range of topics including politics, current events and entertainment.
- Kaggle dataset of *HuffPost*'s news categories, comprising approximately 210,000 real news headlines published between 2012 and 2022.

This combination allowed us to capture a broad spectrum of linguistic styles, ranging from clearly factual reporting to highly exaggerated or ironic phrasing. In particular, the inclusion of *r/NotTheOnion* data introduced challenging *borderline cases*, where genuine news headlines exhibit characteristics similar to satire.

3.2 Data Preprocessing and Cleaning

To ensure consistency across multiple heterogeneous datasets, we first standardized all data into a unified schema consisting of the following fields: Index, Article_Link, Headline, *Is_Sarcastic* (label), and Dataset source. This allowed us to consolidate all entries into a single *master dataset* while preserving traceability to their origins.

Initial preprocessing involved cleaning HTML artifacts and normalizing Unicode characters to ensure textual consistency. We then performed deduplication using both article links and headline text as keys, ensuring that redundant entries were removed.

During preprocessing, we also identified several inconsistencies and noise within the datasets. Notably, some headlines sourced from the Kaggle dataset with *The Onion*'s satirical headlines were found to be non-sarcastic or based on real-world events. This introduced label noise in the form of false positives for sarcasm, which could negatively impact model performance. As such, these headlines were removed from the master copy. The completed data cleanup forms our *master copy deduplicated dataset* (Appendix A).

3.3 Data Addition Strategy

The data addition strategy is driven by the hypothesis that misclassified samples can be categorized into six distinct types:

- **Fuzzy Boundary**: Inherently ambiguous cases where semantically similar examples have conflicting labels. To avoid reinforcing noise, we treat these as irreducible errors or candidates for relabeling rather than adding more data.
- **Outlier**: Samples in underrepresented regions of the feature space. This is a prime target for data addition, as incorporating semantically similar examples can significantly improve generalization.
- **Systematic Bias**: Errors caused by learned spurious correlations, such as associating specific topics with sarcasm. This is mitigated by introducing targeted counter-examples to break these patterns and improve model robustness.
- **Low-Signal Ambiguity**: Short or vague headlines lacking sufficient semantic cues. While new data might help marginally, performance here remains inherently constrained by the lack of contextual information.
- **Punctuation-Based Spurious Correlation**: This occurs when models over-rely on punctuation features such as exclamation marks or question marks, and can be mitigated by adding counter-examples.
- **Mixed Neighbourhood**: Regions in the embedding space where sarcastic and non-sarcastic examples are highly intermixed. Augmenting with more labeled samples in these regions can help the model learn sharper decision boundaries.

Using these categories, we selectively augmented the training dataset by sampling additional examples from the *deduplicated master dataset* while ensuring no overlap with the evaluation data. This controlled data addition approach allowed us to target specific weaknesses without introducing excessive noise.

3.4 Error Analysis and Visualization

To evaluate model behaviour under controlled conditions, we constructed a diagnostic validation set by randomly sampling from a pragmatically filtered pool of our deduplicated master dataset. Specifically, we used Claude Haiku to filter out headlines requiring extralinguistic or world knowledge based on predefined criteria. This ensures the remaining examples are linguistically interpretable without external context, allowing for a focused evaluation of sarcasm detection. However, because this pipeline relies on LLM judgments rather than a human-validated gold standard, it remains an approximation. The filter may inadvertently include headlines that require subtle world knowledge or exclude those where Haiku misses the underlying linguistic cues (prompt in Appendix A).

Additionally, to better understand model behaviour and guide iterative improvements, we developed a data analysis dashboard for systematic error inspection. This dashboard enables cluster-based analysis of model embeddings and prediction errors. Our diagnostic validation results should be interpreted accordingly: they characterise performance on a subset that one capable LLM considers text-only solvable, which is a principled but imperfect approximation of the true boundary.

A key component of the visualization pipeline is the use of *t-SNE* (t-distributed Stochastic Neighbour Embedding) for dimensionality reduction (van der Maaten and Hinton, 2008). Given that each headline is represented as a 768-dimensional embedding from transformer-based models, direct visualization is not feasible. As such, t-SNE projects these high-dimensional embeddings into a 2D space while preserving local neighbourhood structure, providing an intuitive spatial representation of semantic similarity, allowing cluster structure and separation patterns to be examined effectively.

- **Root Cause Distribution Chart**: A root cause distribution chart is constructed to summarize the frequency of errors across the six predefined diagnostic categories defined in the data addition framework. This provides a high-level overview of model weaknesses and supports prioritization of improvement strategies discussed above.
- **t-SNE Side-by-Side Projections**: To further investigate spatial error patterns, we use a dual-panel t-SNE visualization which provides a view of the full embedding manifold without labels, offering a global visualization of the dataset's structure, alongside model errors, where false positives and false negatives are distinguished. This visualization allows us to observe whether errors are randomly dis-

tributed or concentrated in specific regions of the embedding space. Clusters of errors suggest systematic weaknesses in particular semantic regions, while dispersed errors indicate more heterogeneous failure modes.

- **Meso-Level Cluster Analysis**: To bridge the gap between individual errors and global trends, clustering on error embeddings using K-Means where k=8 is also performed in the original 768-dimensional space. Each cluster is then automatically interpreted using TF-IDF-based keyword extraction from member headlines, enabling semantic labeling of clusters. This process reveals recurring thematic patterns in failure cases, which often correspond to systematic misclassification behaviours and shared linguistic structures.

3.5 Methods

We approached sarcasm detection as a supervised binary classification task. Given the short and context-limited nature of headlines, our methods focused on capturing both lexical and semantic features. We experimented with transformer-based models such as *BERT*, *RoBERTa* and *DeBERTa* (Appendix A). These models, which are pre-trained on large corpora, are capable of encoding deeper linguistic patterns, making them well-suited for detecting implicit meanings such as sarcasm. Fine-tuning was performed on our curated dataset to adapt these models to the specific task.

While these models share a common transformer architecture, they differ in their training strategies and representation capabilities. *BERT* (Bidirectional Encoder Representations from Transformers) is trained using masked language modeling and next sentence prediction, enabling it to capture bidirectional context in text (Devlin et al., 2019). *RoBERTa* builds upon *BERT* by removing the next sentence prediction objective, training on larger datasets with dynamic masking and longer training durations, leading to stronger performance and more robust representations (Liu et al., 2019). *DeBERTa* (Decoding-enhanced BERT with disentangled attention) further improves upon this by introducing a disentangled attention mechanism that separates content and positional information, allowing the model to better capture relationships between words (He et al., 2021).

Ultimately, we hypothesise that the results from the prior study on sarcasm detection in social media texts mentioned in related works will generalise to news headlines as well, given that both domains consist of similarly short texts with *implicit meaning* and *contextual ambiguity*. In particular, we expect DeBERTa to perform strongly in this setting due to its architectural improvements, which better model subtle interactions between word content and positional information, an aspect that is especially important for capturing *subtle compositional cues* in sarcastic expressions.

4 Experiments

We analyze model performance by examining the impact of successive modifications to the training pipeline, namely (i) fine-tuning on the original dataset, (ii) hyperparameter tuning, and (iii) targeted data addition. Results are reported against the diagnostic validation dataset unless otherwise stated, as the dataset is constructed to evaluate *sarcasm detection* under a controlled setting where all examples are linguistically interpretable without reliance on extralinguistic knowledge. We report standard classification metrics, including accuracy, precision, recall, and F1-score alongside confusion matrix statistics (TP, FP, TN, FN). These metrics allow us to assess both overall performance and the balance between false positives and false negatives. For clarity and conciseness, we primarily present F1-score in the results, as it provides a balanced summary of precision and recall in the presence of class imbalance. Additional detailed statistics and graphical visualizations are provided in Appendix B.

4.1 Baseline

Among the three transformer models under test, *BERT* serves as a natural starting point for our experiments due to its foundational role in transformer-based language modeling. Related work has shown that *BERT*'s bidirectional contextual representations are particularly effective at capturing *semantic incongruity*, where a statement's literal content contrasts with its intended meaning, a key signal in sarcasm detection (Potamias et al., 2020). As such, we select *BERT* as a zero-shot baseline, without fine-tuning on the given dataset, in order to assess how well its representations, learned from large-scale pre-training corpora, generalize to sarcasm detection in news headlines.

As shown in Table 1, the pretrained *BERT* model performs poorly on the original dataset with an F1-score of 0.5852, as it approximates sarcasm us-

Model	Original	Master	Diag. Val.
BERT	0.5852	0.2562	0.4409

Table 1: Baseline *BERT* F1-score across evaluation datasets.

ing sentiment mismatch and statistical patterns of "news-like vs. unusual text" rather than sarcasm labels. Performance drops significantly on the diagnostic validation set (F1 = 0.4409), as that set is constructed to remove world-knowledge-dependent and shortcut-based signals, eliminating the surface patterns that pretrained *BERT* relies on. This indicates that *sarcasm detection* cannot be solved through general-purpose language representations alone.

4.2 Performance Evaluation

Transitioning from a pretrained zero-shot *BERT* model to one fine-tuned on the original dataset yields a substantial improvement in performance, as shown in Table 2, highlighting the presence of task-relevant linguistic patterns.

Hyperparameter tuning was performed for all three models using *Optuna* (Akiba et al., 2019). Across architectures, we tuned the core optimization and regularization hyperparameters most relevant to stable transformer fine-tuning, including learning rate, warmup, weight decay, training duration, and related model-specific settings. For *BERT*, this included additional contrastive and class-balancing hyperparameters (Khosla et al., 2021; Cui et al., 2019). For *DeBERTa*, fine-tuning was performed using LoRA (Hu et al., 2021), and Optuna jointly tuned both the standard training hyperparameters and the LoRA-specific parameters, namely rank r, scaling factor α, and dropout, applied to the query, key, value, and dense projection layers. These tuning choices were motivated by established findings on transformer fine-tuning and generalization, including the importance of learning rate, warmup, weight decay, layer-wise learning rate decay, and label smoothing (Sun et al., 2020; Loshchilov and Hutter, 2017, 2019; Howard and Ruder, 2018; Szegedy et al., 2015). Across all models, tuning consistently improved F1-score over the corresponding fine-tuned baseline by approximately 0.01 to 0.03. Full search spaces, model-specific tuning dimensions, and selected values for each architecture are reported in Appendix B.

Notably, RoBERTa and BERT experienced immediate F1-score degradation after the first round of data addition (R1 in Table 2). We hypothesise that this reflects their entangled representations: the addition strategy introduces counter-examples whose structural patterns diverge from the original training distribution (§3.3), and an entangled model must reconcile the new content and its positional framing jointly, likely destabilising previously learned associations.

In contrast, DeBERTa absorbed the first two rounds with neutral or positive effects. This robustness is consistent with DeBERTa's disentangled attention mechanism, which separates content and positional information into distinct streams. A disentangled representation can update content-level associations, such as incorporating new semantic boundaries, without simultaneously perturbing positional ones, making it comparatively more tolerant of targeted distributional shifts. DeBERTa's eventual degradation at R3 likely reflects a volume threshold: the added data became large enough to shift the overall training distribution rather than simply extend it, distinct from the immediate noise-sensitivity failure observed in BERT and RoBERTa.

Several limitations of this data addition approach are worth noting. The categorization of misclassified samples involved subjective judgment, and boundary cases between categories were not always clear-cut. The addition pipeline was also constrained by the composition of the master dataset, limiting how much targeted addition was possible for underrepresented categories. Finally, aggregate evaluation scores can mask trade-offs between misclassification types, making it difficult to attribute performance changes to specific addition decisions. These factors suggest the results should be interpreted as indicative rather than definitive.

DeBERTa achieved the highest diagnostic validation F1 at 0.8948, a lead that persisted even after BERT was augmented with *supervised contrastive learning*, *topic-balanced sampling*, *layer-wise learning rate decay* and *label smoothing* (Appendix B). This pattern is consistent with the hypothesis in §3.5 that DeBERTa's disentangled attention mechanism offers representational advantages for the compositional cues characteristic of sarcasm, which training-recipe improvements on a weaker architecture do not recover.

Model	Fine-tuning	Hyp. Tuning	Data Add. R1	Data Add. R2	Data Add. R3
BERT	0.8430	0.8737	0.8611	—	—
DeBERTa	0.8804	0.8948	0.8860	0.8949	0.8795
RoBERTa	0.8632	0.8771	0.8679	0.8111	0.8213

Table 2: Comparison of model F1-score on the diagnostic validation dataset across pipeline stages.

4.3 Structural Error Analysis

To investigate the diminishing returns of our initial unconstrained data addition, we conducted a structural-level analysis on the highest-performing baseline model (*DeBERTa*). We examined which specific linguistic features were systematically over-represented in misclassified samples. Rather than categorizing errors by embedding-space properties, we compared structural features, including first-person voice, question form, second-person voice, and sports terminology.

This analysis revealed two categories exhibiting near-total failure:

- First-person voice: Correctly classified only 27.3% of the time.
- Question-form: 100% of sarcastic question-form headlines were misclassified.

These were not borderline predictions; first-person errors had a mean confidence of 0.944. These patterns traced directly to a structural imbalance in the training corpus. The non-sarcastic class is drawn predominantly from *HuffPost*, where first-person voice and question-based framing are characteristic of opinion columns. Conversely, sarcastic headlines in the training set are disproportionately written in *The Onion's* neutral, third-person news-reporting style. The model had learned these stylistic markers as class-predictive shortcuts rather than recognizing them as neutral with respect to sarcastic intent.

4.4 Targeted Additions

Having established *DeBERTa* as the strongest baseline (F1 = 0.8948), we scoped our targeted structural intervention exclusively to this architecture. We executed a highly targeted structural data addition, adding 610 specific sarcastic headlines surgically targeted to the underrepresented structural categories: 230 first-person, 120 question-form, 100 second-person, 100 sports-domain, and 60 dark-topic headlines that were all extracted from the master dataset. All candidates were filtered to exclude overlap with both the training and evaluation sets using normalised exact matching and fuzzy matching.

Metric	Before	After	Improvement
F1	89.5%	92.5%	3.0%
Accuracy	93.2%	95.3%	2.1%
Precision	93.9%	95.0%	1.1%
Recall	85.5%	90.5%	5.0%

Table 3: Performance metrics before and after targeted data addition.

To assess whether the addition addressed the intended failure modes, we re-evaluated the error rates within each structural category:

Structural Error	Before	After	Reduction
First-person Voice	76.9%	34.6%	∼55%
Question Form	100.0%	33.3%	∼67%
Sports	23.9%	15.0%	∼37%

Table 4: Per-category error rates before and after targeted data addition.

The targeted addition yielded a 3% absolute improvement in F1-score and substantially reduced error rates in the most problematic categories, confirming that the earlier plateau was driven by the composition of added data rather than a fundamental model capacity limit. The disproportionate gain in recall (5.0%) relative to precision (1.1%) further supports this interpretation: the model systematically failed to recognize sarcastic headlines when they were written in formats it associated with factual news, and providing counter-examples in those specific structural categories corrected the bias without introducing new false positives.

However, the residual errors after this intervention are qualitatively distinct from those resolved. The remaining false negatives are concentrated in deadpan, news-like headlines that contain no structural or lexical markers of sarcasm, while the remaining false positives consist of genuine headlines whose content is sufficiently unusual to trigger the model's sarcasm detector despite being factual. These residual cases represent the practical ceil-

ing of a text-only sarcasm detector on this dataset — instances where the distinction between satire and reality cannot be resolved from the headline text alone, aligning with the broader finding discussed in RQ2 that certain forms of sarcasm are fundamentally inaccessible without extralinguistic context.

5 Discussion

RQ1: What error types persist across models, and what do they reveal about model limitations?

Across all models and training configurations, we observe that errors are not randomly distributed but instead form highly consistent and interpretable semantic clusters. This indicates that sarcasm detection performance is primarily constrained by *systematic linguistic biases* rather than model instability. We focus here on the most prominent clusters; additional error patterns are detailed in Appendix C.

A notable observation from the error clusters is that all models consistently classify headlines containing the term "researcher" reliably, but other scientific terms such as "discover", "study" and "scientists" frequently appear in false positive clusters. While the words are semantically similar, these terms differ in distributional usage across the dataset. In particular, terms like "scientists" and "study" occur disproportionately in Onion-style satirical headlines. As such, one might conclude that the effect is thus a *dataset-induced spurious correlation*, with specific keywords becoming a proxy for sarcasm signals. However, even after tuning and data addition, this bias persists, suggesting another possible cause. Further exploration revealed that their usage in sarcastic headlines is due to their capability in producing a stereotypical framing of the headline such as "Scientists discover…" or "Study discovers that…". These constructions invoke an institutional, archetypal authority figure that serves as a setup for the subsequent sarcastic claim, thereby functioning as a common parody template for exaggerated or absurd conclusions. As such, these terms act as strong surface cues for sarcasm during training. In contrast, "researchers" appear more frequently in formal or neutral academic reporting contexts, and is less strongly associated with stereotypical satirical phrasing. It is typically used in conjunction with specific research domains, such as medical or scientific investigations, which reinforces its grounding in legitimate empirical reporting rather than parody-oriented framing.

This raises an important design consideration, whether the association between scientific lexical items and sarcasm constitutes a spurious correlation or a *legitimate stylistic signal* of satirical writing. While the over-representation of phrases does suggest dataset-induced shortcut learning, these constructions also represent consistent rhetorical templates used in satirical journalism. As such, we do not explicitly correct this bias, but interpret it as an *inherent ambiguity* in distinguishing satirical patterns.

Meanwhile, the most robust false negative clusters appear in political satire headlines, particularly those involving entities such as "Trump", "candidates", "Republicans" and politically charged events such as "Hurricane responses". Notably, this effect is largely concentrated in US political discourse, whereas headlines involving figures such as "Putin" or "Xi Jinping" are more consistently classified correctly. This discrepancy may arise from a combination of factors. For one, there is a strong overlap between satirical and non-satirical content across political domains in the dataset, where both genuine news headlines and Onion-style satire frequently share highly realistic, news-like phrasing with minimal lexical distortion, which may make political headlines inherently difficult to separate.

However, this ambiguity is amplified in US political content due to a higher representation in both satirical and non-satirical sources, leading to a denser mixture of positive and negative examples within the same topical space. In contrast, non-US political content is less frequently represented in satirical form, resulting in clearer separation despite similar linguistic surface structures. Another contributing factor is that certain US political headlines in real-world news corpora already contain hyperbolic or seemingly absurd framing. This further blurs the boundary between factual reporting and satirical content. As a result, the model may implicitly learn that such exaggerated framing is compatible with factual news in US political domain, leading it to misclassify satirical headlines as non-sarcastic headlines. Therefore, political sarcasm remains one of the more challenging classes due to *high lexical and syntactic overlap* between satire and genuine reporting.

Notably, not all sarcasm in headlines is expressed through lexical or contextual distortion;

some artifacts exhibit *tautological sarcasm*, where the humorous effect arises from self-evident statements. For example, the headline "Investigators determine Air France disaster caused by plane crash" is syntactically and semantically well-formed as a factual news report, yet is sarcastic due to its redundant framing. Such cases prove particularly challenging for the models, as they lack strong lexical indicators of irony or sentiment shift, and closely resemble legitimate news reporting structures, leading to their frequent misclassification as non-sarcastic content.

RQ2: To what extent can sarcasm be detected using text alone?

A key limitation identified in our analysis is the boundary between *linguistic sarcasm* and *world-knowledge-dependent sarcasm*. Models perform well on explicit linguistic signals like hyperbole or tonal incongruity, but fail when the sarcasm relies on external context. In particular, headlines invoking public figures, cultural references or well-known controversies often require specific background knowledge to interpret the intended irony. For example, a headline such as "Tim Duncan Claims He Has Helped Over 20,000 Women With Sensible Retirement Planning" is syntactically and semantically well-formed as a factual news headline. As such, its sarcasm is only recoverable by a reader who recognizes Tim Duncan's widely known reputation as an unassuming, reserved, and often humorously "unexciting" public figure. This contextual knowledge — *central to the joke* — is entirely absent from the text itself, making sarcasm effectively invisible to a text-only model.

This finding prompted us to refine the scope of the task to focus on *linguistically detectable sarcasm*. As such, headlines that require extralinguistic knowledge, such as specific biographical, cultural or event-level context, or missing visual cues, were excluded. It is of note that the presence of named entities or public figures such as Donald Trump does not by itself trigger exclusion; rather, filtering is applied only when sarcasm cannot be reasonably inferred. This design choice was motivated by a need to ensure a *controlled evaluation setting* in which model predictions can be meaningfully evaluated based on linguistic evidence, rather than external context that is unavailable to text-only models. This ensures that we are better able to isolate and analyse *linguistic failure modes*. Importantly, the original training dataset was left unfiltered to preserve its natural distribution, ensuring that the model still learns from real-world data variability while being evaluated under a more constrained and interpretable problem definition.

This scoping decision reflects the *inherent constraints of a text-only paradigm* rather than a limitation of the model itself. To address headlines requiring external context, future research could employ retrieval-augmented generation (RAG) to dynamically fetch relevant background knowledge, such as recent news events or biographical facts about named entities, providing the model with the real-world grounding necessary to detect situational sarcasm. Alongside knowledge-augmented approaches, other promising directions include leveraging larger generative models with richer parametric knowledge, or developing multimodal frameworks that incorporate images, source metadata, and audience reaction signals. Ultimately, these advancements would transition sarcasm detection from a purely linguistic task into a broader reasoning problem that integrates world knowledge and contextual cues.

6 Conclusion

In this work, we investigated sarcasm detection in news headlines. While fine-tuning, hyperparameter tuning, and targeted data addition, such as our structural interventions for underrepresented phrasing, improved performance, the gains were ultimately *bounded and sensitive to data quality*. Error analysis revealed that failures are driven by systematic linguistic patterns, reflecting both dataset-induced biases and inherent ambiguity in headline construction.

A key limitation is that many forms of sarcasm cannot be resolved through linguistic signals alone. Cases involving political context, public figures, or implicit world knowledge remain challenging, highlighting the hard constraints of a purely text-based formulation. As explored in our analysis, future work should explore *knowledge-augmented or multimodal approaches*, such as dynamically retrieving situational facts or missing visual cues, moving beyond linguistic cues towards incorporating the contextual and pragmatic reasoning required for more robust detection. Ultimately, sarcasm detection may be better framed as a *broader reasoning problem* rather than a purely linguistic classification task.

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Statement of Independent Work

1A. Declaration of Original Work. By entering our Student IDs below, we certify that we completed our assignment independently of all others (except where sanctioned during in-class sessions), obeying the class policy outlined in the introductory lecture. In particular, we are allowed to discuss the problems and solutions in this assignment, but have waited at least 30 minutes by doing other activities unrelated to class before attempting to complete or modify our answers as per the class policy.

To be explicitly clear, AI use is allowed and encouraged for all assignments in CS4248. We have documented our use of AI tools below, as suggested in the NUS AI Tools policy[1].

Documented AI use.

- **Ideation and literature scoping**: Gemini 3 Pro Deep Research and ChatGPT Deep Research were used to surface candidate models, techniques, and related work. All suggested references were independently verified before citation.
- **Methodological suggestions**: Claude was used to suggest papers on fine-tuning techniques, training strategies, and related methods to explore.
- **EDA and coding**: Claude Opus 4.6 and Gemini 3 Pro (via Google Antigravity) were used as coding assistants for data preprocessing, visualizations, and diagnostic tooling.
- **Diagnostic validation filtering**: Claude Haiku 4.5 was used as the linguistic filter described in Section 3.4, with the prompt documented in Appendix A.
- **Writing assistance**: ChatGPT was used for minor stylistic suggestions such as italicisation of technical terms.

Signed, A0271934Y, A0273436B, A0277540B, A0273402N, A0300661U

[1] https://libguides.nus.edu.sg/new2nus/acadintegrity, tab "AI Tools: Guidelines on Use in Academic Work"

A Appendix A

Public repository. The project repository for the experimental code and generated artefacts is https://github.com/MakiseKurisuX/group-18-cs4248. This repository is the canonical location for the scripts, notebooks, and logged outputs used to produce the results reported in this paper.

Repository contents. The repository already contains the versioned datasets needed for our experiments, including the original project dataset, the augmented training dataset variants, the *master copy deduplicated dataset*, and the *diagnostic validation set*. It also contains model-specific README files describing environment setup, training, evaluation, augmentation, and error-diagnostic workflows for the BERT, RoBERTa, and DeBERTa pipelines.

Reproducibility summary. At a high level, reproducing the reported experiments requires the following sequence:

- use the repository's versioned datasets directly, rather than reconstructing them from scratch: this includes the *master copy deduplicated dataset* and the *diagnostic validation set* used for the main controlled evaluation;
- follow the model-specific README instructions already provided in the repository to create the required environments and run the training and evaluation commands for each family;
- for BERT, use the documented Conda environment and the unified `train.py`, `evaluate.py`, calibration, augmentation, and diagnostic scripts under the baseline BERT pipeline;
- for DeBERTa, use the documented `run_sarcasm.py` pipeline with LoRA and Optuna, then run the accompanying error-diagnostic workflow over the saved predictions and checkpoints;
- for RoBERTa, use the documented virtual-environment setup together with the baseline, improved, Optuna-tuned, contextual-augmentation, and augmentation scripts described in that model's README;
- regenerate the evaluation tables, confusion matrices, root-cause summaries, and t-SNE visualisations reported in Appendices B–D from the repository's saved predictions, metrics, and diagnostic outputs.

Claude Haiku Prompt You are an expert linguistic data cleaner for a machine learning dataset. You will be given a list of satirical news headlines. Your job is to determine if the sarcasm can be detected purely through linguistic structure (KEEP) or if it requires outside human context to understand (DROP).

Output ONLY 'DROP' if the headline meets ANY of these conditions:

1. World Knowledge: The joke strictly requires knowing the history, reputation, or current events surrounding specific celebrities, politicians, brands, or sports teams to be funny.

2. Perfect Mimicry (No Absurdity): The text reads exactly like a boring, realistic, 100% normal news alert (e.g., 'senate passes new tax bill'). If there is no linguistic exaggeration, drop it.

3. Missing Visual Context: The text reads like a neutral caption for a photograph we cannot see (e.g., 'the new poster for the movie', 'look at this guy').

4. Cultural Irony: The humor relies on unspoken societal norms, lived experiences, or stereotypes rather than structural wordplay.

Output ONLY 'KEEP' if the headline contains clear linguistic signals of sarcasm:

1. Structural Absurdity: It uses extreme hyperbole, oxymorons, or physically impossible scenarios. (Note: The Absurdity Override - Even if a headline contains a real celebrity name, if the action they are doing is completely absurd or impossible, KEEP it).

2. Tone Clash: It pairs highly formal/academic language with ridiculous, gross, or mundane concepts.

3. The 'Area Man' Trope: It uses fake, generic placeholders (like 'Area Man', 'Nation's Dogs', 'Local Mom') in exaggerated, highly descriptive situations.

Output the results matching the requested JSON schema.

Asset	Role in the pipeline	Location
Course dataset	Original sarcasm benchmark used for fine-tuning and held-out evaluation	Provided as part of the CS4248 project materials
Onion News dataset	Supplemental satirical/non-satirical source used in corpus construction	https://huggingface.co/datasets/Biddls/Onion_News
r/NotTheOnion Kaggle dataset	Borderline real-news source used to expand difficult non-sarcastic cases	https://www.kaggle.com/datasets/thedevastator/discovering-fact-through-humor-investigating-r-n
Onion satire Kaggle dataset	Supplemental Onion headlines used during corpus expansion and cleaning	https://www.kaggle.com/datasets/varunkumarmeda/the-onion-satirical-news-articles
HuffPost News Category dataset	Large non-sarcastic headline source used in corpus construction	https://www.kaggle.com/datasets/zubairdhuddi/news-category-dataset
Public code repository	Training scripts, preprocessing pipeline, evaluation outputs, and supporting artefacts	https://github.com/MakiseKurisuX/group-18-cs4248

Table 5: Dataset and resource inventory for corpus construction and reproducibility.

B Appendix B

B.1 BERT Statistics

Mode	Original F1	Diagnostic Val. F1
pretrained	0.5852	0.4409
orig_no_tune	0.9146	0.8510
orig_no_tune_cal	0.9186	0.8501
orig_tuned	0.9256	0.8615
orig_tuned_cal	0.9273	0.8633
imp_no_tune	0.9149	0.8395
imp_no_tune_cal	0.9159	0.8430
imp_tuned	0.9270	0.8737 (best)
imp_tuned_cal	0.9255	0.8723
imp_large_tuned	0.9327 (best)	0.8707
aug_tuned	0.9323	0.8633
master_no_tune*	0.9243	0.9572*

Table 6: BERT pipeline comparison across evaluation datasets.

Cells marked "(best)" denote the strongest comparable `bert-base` result in that column.

* `master_no_tune`'s Diagnostic Validation score is not directly comparable to the held-out settings above. Its training data contains approximately 98.4% of the diagnostic validation set, so that score reflects largely in-distribution performance rather than generalisation.

Column descriptions.

- **Original F1**: F1-score on the original held-out test set (2,851 examples, roughly balanced).
- **Diagnostic Validation F1**: F1-score on the controlled validation set (1,825 examples) after filtering out headlines that require substantial extralinguistic knowledge.
- **`_cal` suffix**: post-hoc decision-threshold calibration on the validation split.

Mode descriptions.

- `pretrained`: zero-shot BERT with no fine-tuning.
- `orig_no_tune`: fine-tuned on the original dataset with default hyperparameters.
- `orig_no_tune_cal`: the same run with post-hoc threshold calibration.
- `orig_tuned`: original pipeline with Optuna tuning over training hyperparameters.
- `orig_tuned_cal`: the tuned original pipeline with post-hoc threshold calibration.
- `imp_no_tune`: original fine-tuning plus supervised contrastive loss and topic-balanced sampling.
- `imp_no_tune_cal`: the same improved untuned run with post-hoc threshold calibration.
- `imp_tuned`: improved pipeline with Optuna tuning, including contrastive-loss parameters and topic-cluster count.
- `imp_tuned_cal`: the tuned improved pipeline with post-hoc threshold calibration.
- `imp_large_tuned`: the tuned improved pipeline using `bert-large-uncased`.
- `aug_tuned`: tuned improved pipeline with 320 additional KNN-retrieved examples from the master dataset.
- `master_no_tune`: improved pipeline trained on the full master dataset and therefore not directly comparable as a held-out result.

Chosen BERT (imp_tuned) breakdown. The tables show the breakdown of our chosen BERT model for comparison with other models such as DeBERTa and RoBERTa.

	Predicted Non-Sarcastic	**Predicted Sarcastic**
Actual Non-Sarcastic	1136 (TN)	76 (FP)
Actual Sarcastic	80 (FN)	533 (TP)

Table 7: Confusion matrix for the chosen BERT model (`improved_with_tuning`).

Metric	**Value**
Accuracy	0.9145
Precision	0.8752
Recall	0.8695
F1-score	0.8723

Table 8: Performance metrics for the chosen BERT model on the diagnostic validation set.

Selected Optuna hyperparameters. The table below records the hyperparameters selected by Optuna for the three tuned BERT modes.

Parameter	`orig_tuned`
Learning rate	4.93×10^{-5}
Batch size	32
Epochs	4
Max length	64
Warmup ratio	0.140
Weight decay	0.0059
LLRD decay	0.950
Grad. accum.	4
Label smoothing	0.124

Table 9: Optuna hyperparameters: original-family tuned mode.

Parameter	`imp_tuned`	`imp_large_tuned`
Learning rate	4.58×10^{-5}	4.73×10^{-5}
Batch size	8	4
Epochs	5	4
Max length	64	64
Warmup ratio	0.053	0.068
Weight decay	0.0384	0.0156
LLRD decay	0.936	0.805
Grad. accum.	2	2
Label smoothing	0.051	0.116
Contrast. weight	0.196	0.097
Contrast. temp.	0.149	0.102
Topic clusters K	6	10

Table 10: Optuna hyperparameters: improved-family tuned modes.

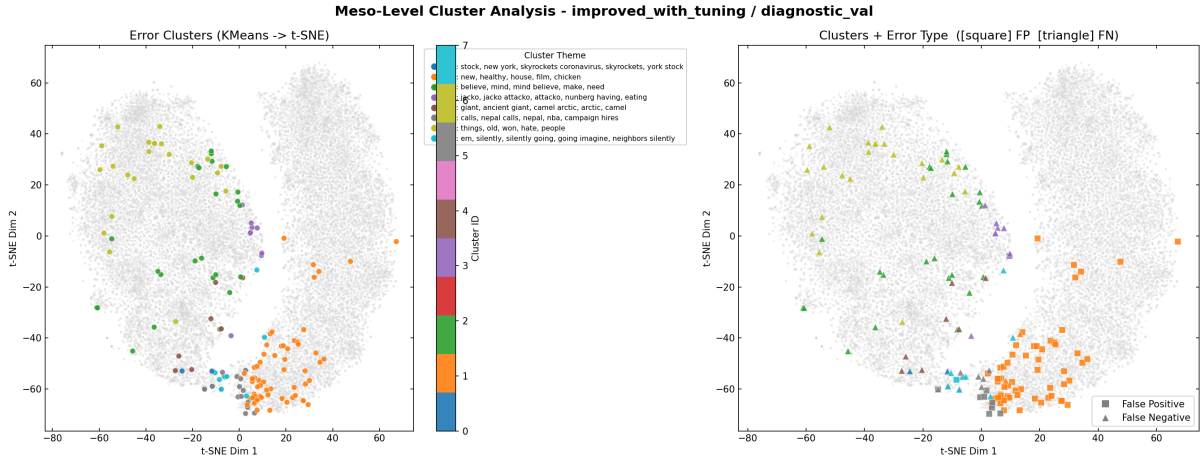

Figure 1: t-SNE visualisation for the chosen BERT model (`improved_with_tuning`); interpretation is provided in Appendix C.

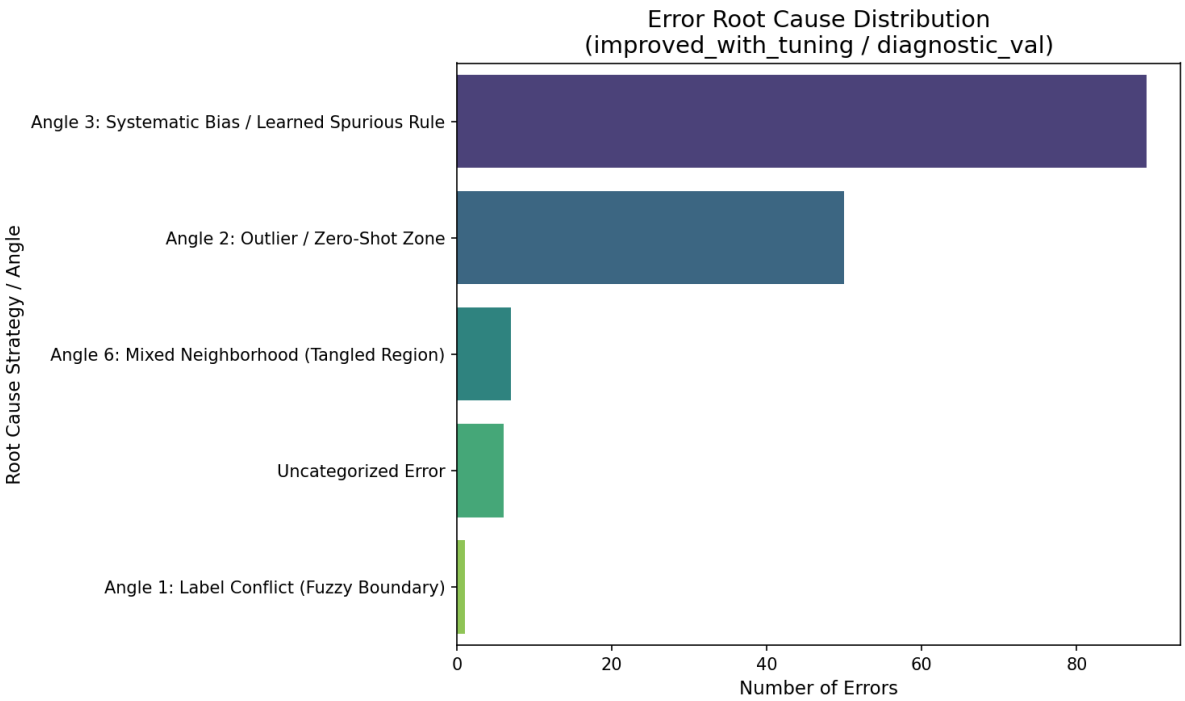

Figure 2: Root cause distribution for the chosen BERT model.

B.2 RoBERTa Statistics

Mode	Original F1	Diagnostic Val. F1
pretrained	0.0000	0.0000
orig_no_tune	0.9620	0.8632
orig_tuned	0.9893 (best)	0.8771 (best)
imp_no_tune	0.9326	0.8282
imp_tuned	0.9537	0.8440
aug_tuned_r1	0.9663	0.8679
aug_tuned_r2	0.9298	0.8111
aug_tuned_r3	0.9026	0.8213

Table 11: RoBERTa pipeline comparison across evaluation datasets.

From the above table, we pick original_with_tuning as the best RoBERTa model. This model variation utilises Optuna Hyperparameter fine-tuning with the roberta-base model.

improved_with_tuning utilises context in addition to the headlines in the Kaggle dataset. The analysis of the methods and results will be further explored in Appendix D.

Mode descriptions.

- `pretrained`: zero-shot baseline with no fine-tuning.
- `orig_no_tune`: fine-tuned on the original dataset with default settings.
- `orig_tuned`: original training pipeline with Optuna tuning.
- `imp_no_tune`: improved pipeline with additional contrastive and balancing components, but no tuning.
- `imp_tuned`: improved pipeline with tuning.
- `aug_tuned_r1`, `aug_tuned_r2`, `aug_tuned_r3`: successive rounds of targeted data addition on top of the tuned pipeline.

Metric	Value
Accuracy	0.9200
Precision	0.9061
Recall	0.8499
F1-score	0.8771

Table 12: Performance metrics for the best RoBERTa model (`original_with_tuning`).

Parameter	`orig_tuned`
Learning rate	1.07×10^{-5}
Batch size	32
Epochs	4
Warmup steps	300
Weight decay	0.259
Label smoothing	0.0895
LR scheduler	cosine

Table 13: Hyperparameters for RoBERTa sarcasm detection

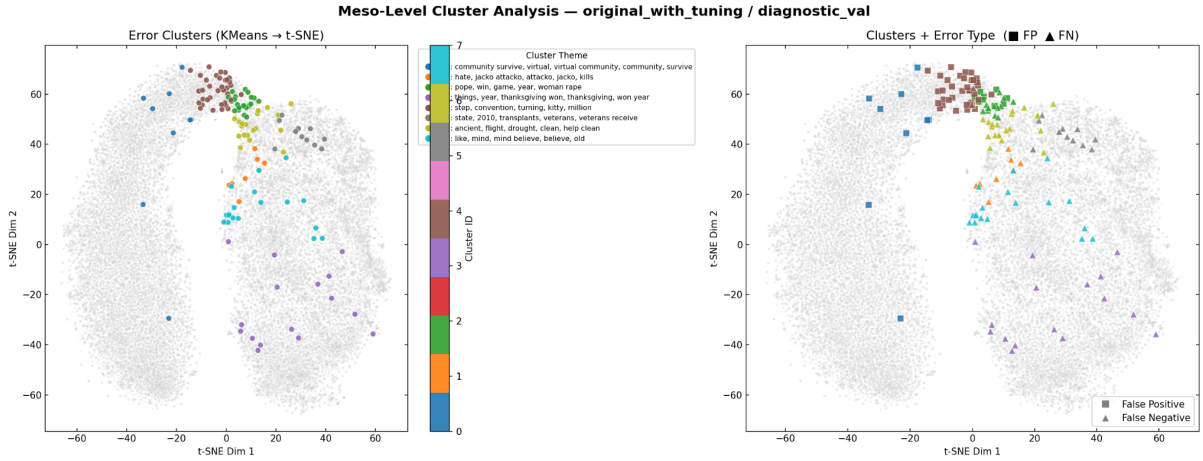

Figure 3: t-SNE visualisation for the best RoBERTa model (`original_with_tuning`); interpretation is provided in Appendix C.

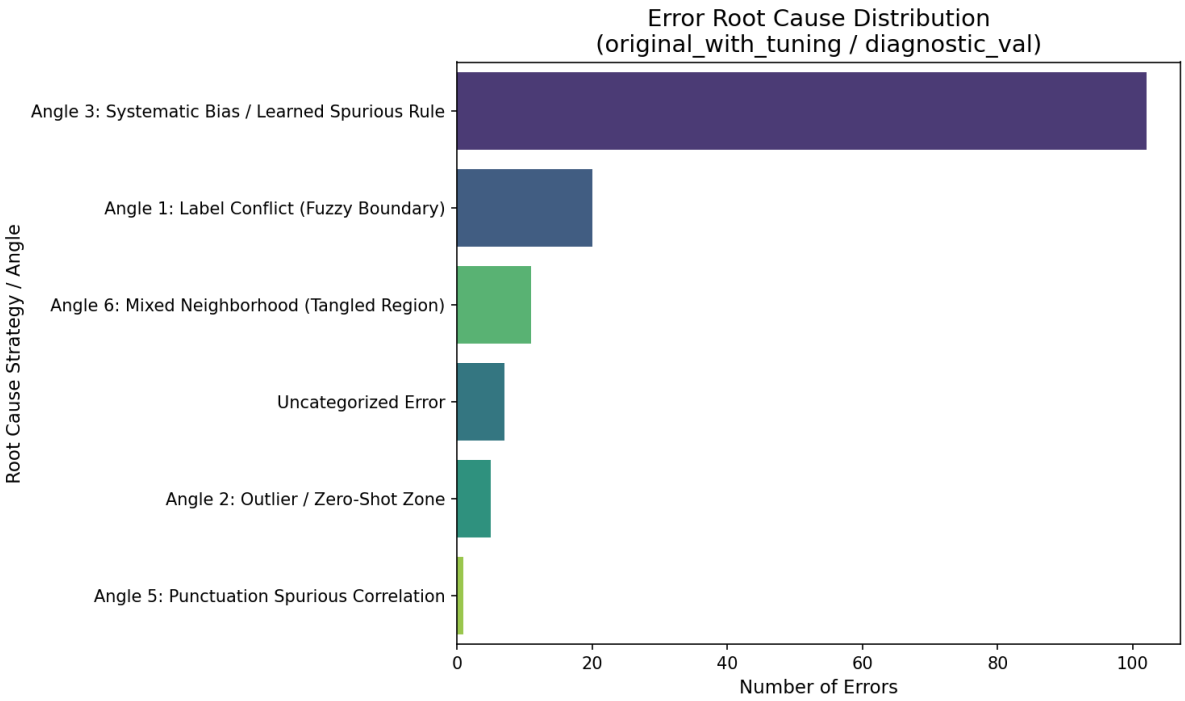

Figure 4: Root cause distribution for the best RoBERTa model.

B.3 DeBERTa Statistics

Mode	Original F1	Diagnostic Val. F1
`pretrained`	0.0000	0.0000
`orig_no_tune`	0.9291	0.8804
`orig_tuned`	0.9373	0.9024
`imp_tuned`	0.9399	0.8893
`aug_tuned_r1`	0.9396	0.8860
`aug_tuned_r2`	0.9402	0.8949
`aug_tuned_r3`	0.9383	0.8795
`aug_new_tuned`	0.9436 (best)	0.9250 (best)

Table 14: DeBERTa pipeline comparison. The final row corresponds to the targeted structural data addition reported in the main paper.

Column descriptions.

- **Original F1**: F1-score on the train-test-split set.
- **Diagnostic Validation F1**: F1-score on the controlled validation set (1,825 examples) after filtering out headlines that require substantial extralinguistic knowledge.

Mode descriptions.

- `pretrained`: zero-shot deBERTa with no fine-tuning.
- `orig_no_tune`: fine-tuned on the original dataset with default hyperparameters.
- `orig_tuned`: original pipeline with Optuna tuning over training hyperparameters.
- `imp_tuned`: improved pipeline with Optuna tuning, including contrastive-loss parameters and topic-cluster count.
- `aug_tuned_r1`: tuned improved pipeline with additional KNN-retrieved examples from the master dataset.
- `aug_tuned_r2`: tuned improved pipeline with additional KNN-retrieved examples from the master dataset.
- `aug_tuned_r3`: tuned improved pipeline with additional KNN-retrieved examples from the master dataset.
- `aug_new_tuned`: improved pipeline trained on the full master dataset and therefore not directly comparable as a held-out result.

Metric	Value
Accuracy	0.9534
Precision	0.9462
Recall	0.9048
F1-score	0.9250

Table 15: Performance metrics for the best DeBERTa model after targeted structural data addition.

Parameter	`deberta_hyperparameters`
Learning rate	4.33×10^{-4}
Batch size	16
Epochs	5
Warmup ratio	0.172
Weight decay	6.95×10^{-4}
Label smoothing	0.141
LoRA rank (r)	16
LoRA alpha (α)	8
LoRA dropout	0.090

Table 16: Hyperparameters for DeBERTa-v3 sarcasm detection (LoRA fine-tuning).

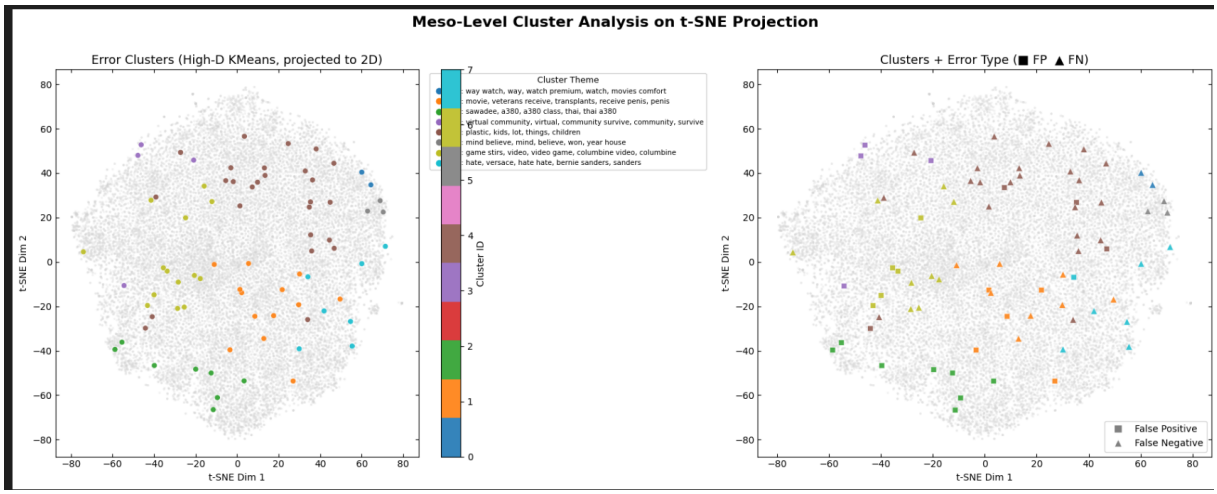

Figure 5: t-SNE visualisation for the final DeBERTa model; interpretation is provided in Appendix C.

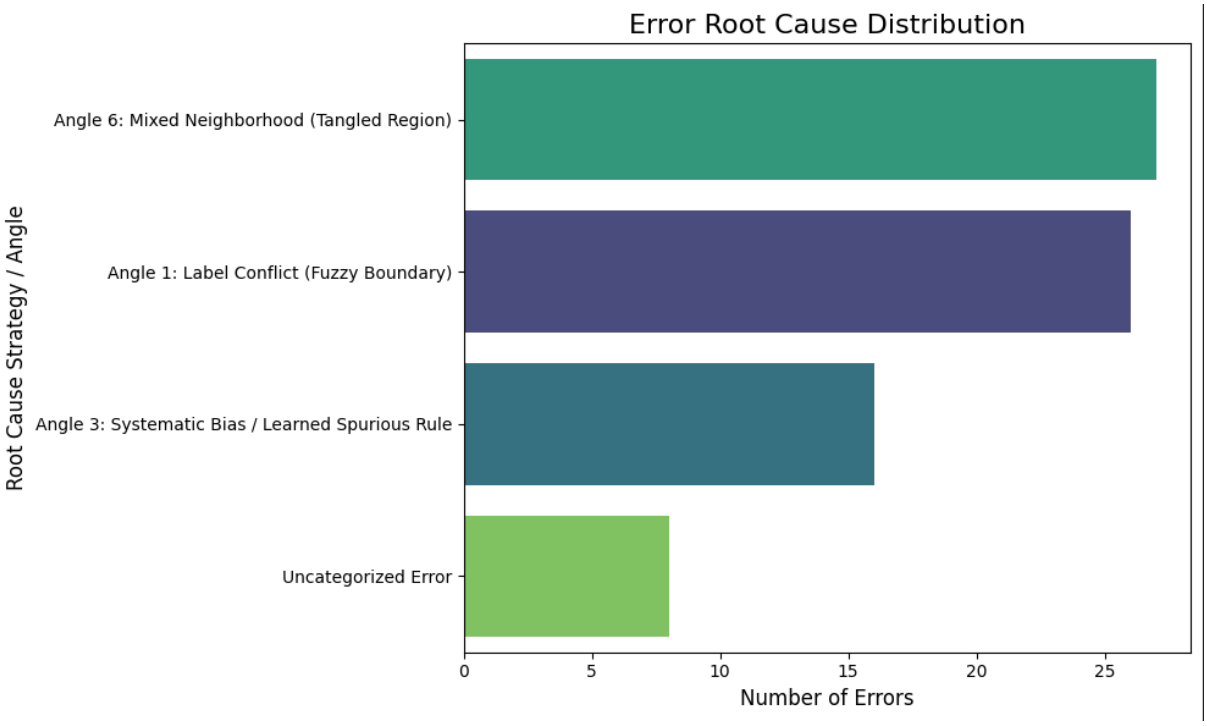

Figure 6: Root cause distribution for the final DeBERTa model.

C Appendix C

C.1 BERT (`improved_with_tuning`) Meso-Level Cluster Analysis

The table below summarises the eight error clusters produced by K-Means ($K = 8$) on BERT CLS-token embeddings from the `improved_with_tuning` model evaluated on the diagnostic validation set (threshold $\tau = 0.50$, 153 total errors: FP = 69, FN = 84). Top cluster keywords are extracted via per-cluster TF-IDF.

Cluster	Top Keywords	Character
C0	stock, skyrockets, coronavirus, york	Financial parody FN
C1	new, healthy, house, film, chicken	General news FP
C2	believe, mind, make, need	Dry/first-person Onion FN
C3	jacko, attacko, nunberg, eating	Absurdist/noise FN
C4	giant, ancient, camel, arctic	Unusual news FN
C5	calls, nepal, nba, campaign	Mixed
C6	things, old, won, hate, people	Dry/sardonic Onion FN
C7	silently, going, imagine, neighbors	Absurdist FN

Table 17: BERT meso-level error clusters: semantic summary.

Cluster	Size	FP	FN
C0	2	0	2
C1	61	59	2
C2	24	0	24
C3	11	1	10
C4	7	0	7
C5	16	8	8
C6	23	0	23
C7	9	1	8

Table 18: BERT meso-level error clusters: error counts.

C0 — Financial/news parody FN. A very small cluster of two Onion headlines using financial and news-ticker framing that the model failed to flag as sarcastic. The representative example "dow skyrockets after coronavirus begins trading on new york stock exchange" parodies financial reporting but reads superficially like a genuine headline.

C1 — General news FP. The dominant error cluster, accounting for 59 of the 69 total false positives. Genuine policy, culture, and everyday news headlines were systematically over-predicted as sarcastic. Representative examples include "california raises smoking age to 21" and "gay couple claims russia accidentally recognized their marriage", both real news items.

C2 — Dry and first-person Onion FN. All 24 errors in this cluster are false negatives. Sarcasm is carried entirely by narrative voice and framing rather than absurd vocabulary. Representative examples include "my great-grandfather came to this country with nothing but $10 in his pocket, $300,000 in his bank account, and a dream" and "time flies when you're feeding an addiction to internet pornography over which you no longer have any control".

C3 — Absurdist/noise FN. Mostly false negatives, with one false positive. The cluster groups absurdist Onion headlines anchored to low-frequency named tokens alongside ambiguous news quotes. Representative examples include "jacko is on the attacko!" and "sam nunberg: 'i'm not having a meltdown"'.

C4 — Unusual news FN. All seven errors are false negatives. Extraordinary-sounding but plausible headlines that the model failed to recognise as satirical. Representative examples include "lobster diver survives being trapped in whale's mouth" and "high schoolers given detention for cutting class during active shooting".

C5 — Mixed. Balanced at 8 FP and 8 FN. No single failure mode dominates. The cluster mixes institutional news headlines with satirical takes on events and campaigns. Representative examples include "hurricane nicole strengthens into dangerous category 4 storm" (FP) and "investigators determine air france disaster caused by plane crash" (FN).

C6 — Dry and sardonic Onion FN. All 23 errors are false negatives. Onion content written in a quiz-show, editorial, or list register where the vocabulary is entirely ordinary. Representative examples include "the pros and cons of letting children die" and "so you think you can win a presidential medal of freedom?".

C7 — Absurdist FN. Mostly false negatives (8 of 9). Deadpan Onion headlines that use mundane or bureaucratic language to deliver absurdist premises. Representative examples include "proposed legislation would require airline seats meet federal ass standards" and "syria conflict intensifies as bears enter war".

C.2 RoBERTa (`original_with_tuning`) Meso-Level Cluster Analysis

Cluster	Top Keywords	Character
C0	community survive, virtual, virtual community, community, survive	Quirky / tonal ambiguity FP
C1	hate, jacko attacko, attacko, jacko, kills	High-absurdity Onion FN
C2	pope, win, game, year, woman rape	Mixed topical/news + satire overlap
C3	things, year, thanksgiving won, thanksgiving, won year	Dark humour / ethical satire FN
C4	step, convention, turning, kitty, million	General news FP
C5	state, 2010, transplants, veterans, veterans receive	Institutional / official-style satire FN
C6	ancient, flight, drought, clean, help clean	Straight-tone Onion FN
C7	like, mind, mind believe, believe, old	Subtle / conversational sarcasm FN

Table 19: RoBERTa meso-level error clusters: semantic summary.

Cluster	Size	FP	FN
C0	9	9	0
C1	9	0	9
C2	23	9	14
C3	16	0	16
C4	38	36	2
C5	11	0	11
C6	21	0	21
C7	19	0	19

Table 20: RoBERTa meso-level error clusters: error counts.

C0 — Quirky / tonal ambiguity FP. These are genuine or only mildly humorous headlines misclassified as sarcastic because of quirky phrasing. Representative examples include "unimpressed cat is not at all interested in your magic trick" and "virtual community: can we survive it?".

C1 — High-absurdity Onion FN. RoBERTa still misses obviously exaggerated Onion headlines in this cluster. Representative examples include "jacko is on the attacko!" and "deal alert: the quantity of raw plastic and silicon that makes up an xbox series x can be yours for just $8.27".

C2 — Mixed topical/news + satire overlap. This broad cluster mixes real news and satire across politics, religion, and daily topics. Representative examples include "36-year-old man begins outlining a savings plan for ps5" and "nancy pelosi sets record for longest continuous speech in house history".

C3 — Dark humour / ethical satire FN. Sarcastic headlines with morally uncomfortable or controversial humour remained difficult. Representative examples include "the pros and cons of letting children die" and "we're investing so much in alternative fuels, sometimes we almost forget to pump oil!".

C4 — General news FP. This is a large false-positive cluster of genuine headlines with slightly unusual phrasing. Representative examples include "queen elizabeth ii has assistants break in her shoes, document all her outfits" and "colorado asks unemployed to pay back $128 million in jobless benefits".

C5 — Institutional / official-style satire FN. RoBERTa struggled with satire framed like official announcements. Representative examples include "yankees to build stadiums in every mlb city for away games" and "tennessee to make public camping for homeless a felony".

C6 — Straight-tone Onion FN. These headlines are sarcastic but written in a neutral, news-like tone. Representative examples include "oil spill engulfs lincoln memorial after u.s. lifts drilling restrictions on national monuments" and "facebook adds more than 50 new gender options".

C7 — Subtle / conversational sarcasm FN. This cluster captures sarcasm expressed through wordplay or conversational tone instead of overt absurdity. Representative examples include "my kids are learning how to drive...me up the wall!" and "i couldn't help but notice your product hasn't been endorsed by anyone yet".

C.3 DeBERTa (`augmented_new_with_tuning`) Meso-Level Cluster Analysis

Cluster	Top Keywords	Character
C0	house, year, class, hate, major	Large mixed/general
C1	plastic, man, surgery, just, deal	Health/medical parody
C2	campaign, hires, rubio, brothers, strategist	Political news
C3	service, royal, maroc, customer, air	Mixed/noise
C4	winter, heat, movie, meetings, indoor	Sports/entertainment parody
C5	old, family, suburb, maintain, ways	Dry/relational parody
C6	things, looking, psychopaths, cats, prove	Media critique / pop-psych
C7	survive, community, virtual, invasion, ranked	Niche/unusual news

Table 21: DeBERTa meso-level error clusters: semantic summary.

Cluster	Size	FP	FN
C0	56	20	36
C1	4	1	3
C2	3	2	1
C3	2	1	1
C4	3	0	3
C5	3	0	3
C6	4	1	3
C7	2	2	0

Table 22: DeBERTa meso-level error clusters: error counts.

C0 — Large mixed/general. This is a large catch-all cluster caused by the high semantic variance of the remaining errors. Representative examples include "job growth remains sluggish, or rather, akin to a slug" and "massachusetts mayors battle melatonin brownie".

C1 — Health/medical parody. The model still misses some sarcastic headlines that borrow the sterile tone of medical reporting. Representative examples include "new children's book helps kids deal with pain and isolation of plastic surgery" and "man's breast friend: who pays for plastic surgery in divorce?".

C2 — Political news. Genuine political campaign headlines can still be pulled toward satire. Representative examples include "rubio campaign hires top koch brothers strategist" and "carson now top fundraiser in iowa, campaign cites ground game".

C3 — Mixed/noise. This small cluster is dominated by TF-IDF artefacts and narrow lexical overlap. Representative examples include "i didn't become a secret service agent for the jimmy carter midnight to 6 a.m. shift" and "royal air maroc + customer service = 0".

C4 — Sports/entertainment parody. These are subtle parody headlines framed like routine sports or events coverage. Representative examples include "winter meetings heat up with rumors that hotel has an indoor pool" and "nhl tries to pass off commercial flight as flyover for winter classic".

C5 — Dry/relational parody. This cluster remains difficult because the sarcasm is deadpan and relational rather than lexically exaggerated. Representative examples include "in this family we maintain the ways of the old suburb" and "packers tell fans they gave favre to a nice farm family".

C6 — Media critique / pop-psych. These headlines mimic listicle and pop-psychology structures that also appear in genuine clickbait. Representative examples include "things all cats do that prove they are psychopaths" and "psychological effects of connecticut shooting: 'you're looking at a lot of trauma' ".

C7 — Niche/unusual news. DeBERTa can still over-predict sarcasm for bizarre but genuine headlines. Representative examples include "2 utah cities ranked among worst to survive dragon invasion" and "virtual community: can we survive it?".

D Appendix D

RoBERTa Improvements

For the *improved_no_tuning* and *improved_with_tuning* RoBERTa models, we utilised the technique proposed by (Helal et al., 2024). The paper explores the influence of contextual information, as it can affect the intended meaning of sarcastic statements.

Headlines are oftentimes short, and might not provide enough context to the model to predict whether or not it is sarcastic. As such, we append article description, section and author name to the headline, with a separator token <s>, as the model's input. For the scraping of such information, we followed robots.txt requirements, and only obtained the article descriptions that we are allowed to scrape. For descriptions that are irrelevant, we left it empty for that particular document. It is also worth mentioning that there were quite a few invalid URLs in the given Kaggle dataset. In total, we managed to scrape 5,045 out of 28,619, which amounts to approximately 17.63% coverage.

Mode	Original F1	Master Copy F1	Diagnostic Validation F1
original_no_tuning	0.9620	0.7360	0.8632
original_with_tuning	0.9893	0.7484	0.8771
improved_no_tuning	0.9326	0.7188	0.8282
improved_with_tuning	0.9537	0.7329	0.8440

Table 23: F1-score comparison across different RoBERTa configurations

From our results, we can see that the improvement actually causes poor results across all 3 evaluation datasets. Hyperparameter fine-tuning does improve the results quite substantially, but it still falls behind a non-fine-tuned roberta-base model.

However, if we compare the percentage improvements for all 3 evaluation datasets, the improved model with hyperparameter tuning actually has a higher improvement for Master Copy F1 (1.68% vs 1.96%) and Diagnostic Validation F1 (1.61% vs 1.91%). While the contextual augmentation may indeed introduce some noise, it may also provide useful signals if we combine it with more training strategies. The quality of the context also matters, which could have been verified better for training. Given that our coverage for the additional data is only 17.63%, we can attribute it to the lack of additional context, in addition to some invalid URLs, that does play a part in the decrease in F1 score across all 3 datasets.

Overall, contextual augmentation heavily relies on the quality as well as coverage of the additional context. Given that our dataset also missed on our article section, there may be further improvements in the F1 score and the raw prediction of the RoBERTa model, if we manage to procure sufficient data.